

RCF Phase C Mathematical Program

Consolidated proof strategies for the six open theorem targets: $D=3$ + triangle inequality (Fubini-Study metric), relational burden ($\hat{F}$ decomposition), T-1/T-2 (spectral gap of $\hat{F}$, unified via Lieb-Robinson), T-4 (Born rule via Petz/Bures geometry), T-6 (entropy-area via channel suppression). Resolves the $-\log$ vs $\arccos$ question: use both, in different roles. Includes ripple-effect trace and lemma-skeleton deep-dive on the Fubini-Study resolution.

6

OPEN TARGETS

2

CLOSEST TO DONE

1

T-1/T-2
UNIFICATION

7

DEEP-DIVE
LEMMAS

AUDIENCE

MATH DEPTH

DATE

Internal Framework Team

Strategy Level + Lemma-Skeleton Appendix

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Chapter 1 — Executive Summary

This document consolidates the brainstorming batch on Phase C mathematical strategies for the Reconciliation Causal Framework. Six reviewer voices proposed approaches to the framework's six open theorem targets: T-1 (γ derivation), T-2 (spectral gap), T-4 (Born rule / Z-invariance), D=3 (spatial dimensionality + triangle inequality), T-6 (entropy-area scaling), and relational burden (dark matter mechanism). The voices converge on several points and disagree on others; this document resolves the disagreements by picking the mathematically correct side.

The convergent verdict: the framework's open targets are not independent — they reduce to a small number of shared mathematical ingredients. The single most important unification is that **T-1 and T-2 both reduce to the spectral gap of $\mathbf{F}$** : the same gap that determines the Convergence Theorem's convergence rate (T-2) also determines the causal speed limit $c = \ell_0 \cdot \sqrt{\lambda_{\min}(\mathbf{F})}$ (T-1). The shared prerequisite for both is an **adjacency structure on the constraint index set $\{\hat{\mathbf{C}}_{\alpha}\}$** , which can be derived from Jordan co-occurrence ($A \circ B$ defines 'who's near whom').

The Two Closest-to-Done Targets

D=3 + Triangle Inequality (Ch 4): the Fubini-Study metric swap ($d_{\omega} = \ell_0 \cdot \arccos(\sqrt{K_{\omega}})$) makes the triangle inequality automatic and kills the Factorization + Dominance apparatus. D=3 then follows from the Gram rank of the Jordan algebra's three independent relational invariants. **Relational Burden** (Ch 5): the decomposition $\mathbf{F} = \mathbf{F}_{\text{intra}} + \mathbf{F}_{\text{cross}}$ makes relational burden = cross-sector linking cost, giving dark matter a computable profile. Both are strategy-level and ready for proof execution.

The user's specific concern — whether to replace $-\log$ with $\arccos$ for the distance function — is addressed in Chapter 3. The resolution: **don't replace one with the other; use both, in different roles**. Fubini-Study ($\arccos$) is the metric (geometry); $-\log(K_{\omega})$ is a potential function (information theory). They are related by differentiation, not by substitution. This is standard in Riemannian geometry and quantum information geometry (Petz 1996).

The document is organized by priority: the two closest-to-done targets get full strategy-level treatment (Ch 4–5); the two strategy-outline targets (T-1, T-2) get proof-strategy outlines (Ch 6–7); the two status-assessment targets (T-4, T-6) get status assessments with circularity/gap flags (Ch 8–9). Chapter 10 traces the downstream ripple effects of each proposed change. Chapter 11 is a deep-dive appendix developing the Fubini-Study resolution to lemma-skeleton level.

Chapter 2 — Methodology

The brainstorming batch contained six distinct reviewer voices. They were consolidated by: (a) reading all proposals for each of the six targets; (b) identifying where voices converge and where they disagree; (c) resolving disagreements by picking the mathematically correct side (with reasoning); (d) classifying each target by readiness (closest-to-done, strategy outline, status assessment); (e) tracing the downstream ripple effects of each proposed change.

2.1 Critical Stance: Resolve Disagreements

Where voices conflict, this document picks the mathematically correct side and explains why. The four key disagreements resolved:

ID	Disagreement	Resolution
D1	T-2 via fixed $D=3$ (several voices) vs 'D=3 doesn't bound Hilbert dim' (one reviewer)	RESOLVED: $D=3$ fixes SPATIAL dimension, not HILBERT-SPACE dimension. Ordinary 3D QM has infinite-dim H . The correct path: $D=3$ fixes the LOCAL ALGEBRA ($M_d(C)$), giving finite-dim local sectors; detectability lemma lifts local gaps to global. The 'dimensions don't scale' intuition is right but the mechanism is different.
D2	'Least resistance -> Born rule' (several voices) vs 'circularity risk' (one reviewer)	RESOLVED: the burden cost B_{decomp} already encodes $ c_i ^2$ via the density matrix. 'Minimizing burden yields Born' is tautological UNLESS burden is defined independently of sector weights. The Petz/Bures-metric direction (derive Born from the metric the dynamics already uses) is more promising.
D3	K_ω not Jordan-built (reviewer R4) vs 'symmetry comes from Jordan structure' (v2.0 Remark 3.1.3)	RESOLVED: R4 is correct. $K_\omega = \omega(A \text{ dagger } B) ^2/\dots$ does NOT use $A \text{ circ } B$. The fix: $K_\omega^J(A,B) = \omega(A \text{ circ } B)/\sqrt{\dots}$ genuinely uses the Jordan product. Fubini-Study still works (arccos on any inner product). The $D=3$ Gram-rank argument then becomes representation-theoretic.
D4	$\gamma = \sqrt{\lambda_{\min}(F_{\text{hat}})}$ (several voices) vs ' $F_{\text{hat}} \rightarrow 0$ on physical sector' (caveat)	RESOLVED: $\lambda_{\min}$ is the gap on the ORTHOGONAL COMPLEMENT of $\ker(F_{\text{hat}})$ (smallest non-zero eigenvalue). This is the same gap T-2 needs. T-1 and T-2 are unified: one spectral gap, two physical consequences (convergence rate + causal speed).

2.2 Math Depth: Strategy Level

Each target is treated at the proof-strategy level: name the mathematical technique (Lieb-Robinson, detectability lemma, Fubini-Study, Petz metric, Gleason), state the theorem to be proved, sketch the derivation in 3–5 steps, and list dependencies. No full formal proofs — except in the deep-dive appendix (Ch 11), which develops the Fubini-Study resolution to lemma-skeleton level.

2.3 Ripple-Effect Tracing

For each proposed change (Fubini-Study swap, $\mathbb{F}$ decomposition, $\gamma = \sqrt{\lambda_{\min}(\mathbb{F})}$, K_{ω^J} Jordan-built, adjacency structure), Chapter 10 traces what changes downstream: what gets killed, what gets redefined, what stays unaffected, what opens up. The Fubini-Study swap in particular changes the appearance of ℓ_0 in several places — but the trace shows that ℓ_0 is algebraic (from spectral discreteness), not metric-dependent, so κ_B and BH entropy survive unchanged.

Chapter 3 — The Metric Question: Fubini-Study vs Log vs Both

The user's remark: "If possible we can find a replacement for $-\log$ or $\arccos$ because both of them fix a problem the other can't and replacing one with the other will just be changing the problem." This is exactly right. The two functions have incompatible properties:

3.1 The Two Functions and Their Incompatible Properties

Function	Formula	Has	Lacks
Log distance (v2.0 current)	$d_\omega = -\ell_0 \cdot \log(K_\omega(A,B))$	Product→sum: $f(xy) = f(x) + f(y)$. When $K_\omega(A,C) = K_\omega(A,B) \cdot K_\omega(B,C)$ (Cubic Factorization, Condition 3.3.1), the distance is exactly additive: $d(A,C) = d(A,B) + d(B,C)$.	NOT an unconditional metric. The triangle inequality holds only when Factorization + Dominance both hold (two extra structural conditions, one of which — Dominance — is flagged as circular in the patch plan).
Fubini-Study distance (proposed swap)	$d_{FS} = \ell_0 \cdot \arccos(\sqrt{K_\omega(A,B)}) = \ell_0 \cdot \arccos(K_\omega(A,B))$	Unconditional metric. $\arccos(\langle \psi \varphi \rangle)$ is the geodesic distance on projective Hilbert space. Triangle inequality is AUTOMATIC — no Factorization, no Dominance, no extra conditions.	No product→sum structure. $\arccos(xy) \neq \arccos(x) + \arccos(y)$ in general. The cubic kernel $K_\omega(A,B,C) = K_\omega(A,B) \cdot K_\omega(B,C)$ does NOT simplify to an additive distance form under $\arccos$.

3.2 The Impossibility

No Single Function Does Both

There is NO single function $f: [0,1] \rightarrow [0,\infty)$ such that $d = \ell_0 \cdot f(K_\omega)$ is BOTH an unconditional metric AND has the product→sum property. The product→sum condition $f(xy) = f(x) + f(y)$ uniquely forces $f = -c \cdot \log$ (the only continuous solution). But $-\log$ is not an unconditional metric (it fails the triangle inequality when factorization fails). Conversely, $\arccos$ is a metric but lacks the product→sum structure. The two properties are mathematically incompatible for a function of K_ω alone.

3.3 The Resolution: Use Both, in Different Roles

Don't replace one with the other. USE BOTH, in different roles. This is standard in Riemannian geometry and quantum information geometry:

- Fubini-Study ($\arccos$) is the METRIC — the geodesic distance on projective Hilbert space. It gives the triangle inequality, the tangent space, the geodesic structure, the Riemannian volume element. It is the primary

geometric object.

- $-\log(K_\omega)$ is a POTENTIAL FUNCTION — the Rényi-2 relative entropy (also called the "logarithmic distance" or "potential of the metric"). Its Hessian gives the metric tensor: $g_{ij} = \partial_i \partial_j (-\log K_\omega)$. It encodes the information-theoretic content and appears in κ_B , BH entropy, and the cubic kernel's multiplicative structure.

The two are related by DIFFERENTIATION, not by substitution. The metric is the Hessian of the potential — this is exactly how Riemannian geometry works (the distance function and the potential function are distinct objects, related by the geodesic equation).

3.4 Why This Addresses the User's Remark

The user's concern — "replacing one with the other will just be changing the problem" — is exactly right IF we try to use one function for both purposes. The resolution is to stop trying. The Fubini-Study metric handles geometry (where metricity is needed); the log potential handles information theory (where the product→sum structure is needed). Neither replaces the other. The cubic kernel $K_\omega(A,B,C) = K_\omega(A,B) \cdot K_\omega(B,C)$ survives because it's a property of the KERNEL (K_ω), not of the distance function — and the kernel is unchanged. The κ_B and BH formulas survive because they use ℓ_0 as the fundamental scale (from the algebra's spectral discreteness), not from the distance function.

3.5 The Deep Point: Petz's Classification

In Petz's 1996 classification of monotone quantum metrics, the Bures metric (which reduces to Fubini-Study for pure states) is one specific member of a family, singled out by extra symmetry requirements. The Rényi relative entropies (including $-\log K$ as the order-2 case) are the POTENTIALS that generate these metrics via differentiation. The framework already uses the Bures metric implicitly (Open Extension is gradient descent on the Bures metric, §1.5.3). Adopting Fubini-Study explicitly as the distance makes the geometry consistent with the dynamics — and the log potential remains available for the information-theoretic formulas that need it.

Chapter 4 — The Six Targets (Strategy-Level Treatment)

Each target is presented with: the problem, the proposed approach, the mathematical technique, the theorem to prove, the derivation sketch (3–5 steps), dependencies, ripple effects, disagreements resolved, and status. Targets are ordered by priority: closest-to-done first (Ch 4–5), then strategy outlines (Ch 6–7), then status assessments (Ch 8–9). Note: Chapter 4 of this document is the D=3 target; the executive summary was Chapter 1, methodology Chapter 2, the metric question Chapter 3.

4.1 [T1] D=3 + Triangle Inequality (CLOSEST TO DONE)

► [T1] D=3 + Triangle Inequality

Priority: CLOSEST TO DONE | v2.0 location: Def 3.1.2 (K_ω), Def 3.2.1 (d_ω), Condition 3.3.1 (Factorization), Condition 3.3.2 (Dominance), Thm 3.6.4 (D=3 closure)

The problem. v2.0's $d_\omega = -\ell_0 \cdot \log(K_\omega)$ is not an unconditional metric — it needs Cubic Factorization (Condition 3.3.1) AND Cubic Dominance (Condition 3.3.2) to force the triangle inequality. Dominance is flagged as circular in the patch plan. The D=3 closure (Thm 3.6.4) was demoted to 'Conditional Proposition' because the closure defect $\Xi_C(D)$ was tautological ($(D=3)$) and the three-channel assumption (existence, position, direction) is ad hoc — not derived from the algebra.

The proposed approach. Two-part fix: (A) METRIC FIX — Swap d_ω from $-\ell_0 \cdot \log(K_\omega)$ to $d_\omega = \ell_0 \cdot \arccos(\sqrt{K_\omega}) = \ell_0 \cdot \arccos(|K_\omega|)$. This is the Fubini-Study distance — the geodesic distance on projective Hilbert space. The triangle inequality becomes AUTOMATIC (geodesic distances are metrics). This kills the Factorization + Dominance apparatus outright (no conditions needed) and makes the tangent space a genuine Riemannian object (directions become real tangent vectors, not ad hoc profile differences). (B) D=3 FROM CHANNEL RANK — The three inference channels (Existence from Quadratic, Distance + Direction from Cubic) correspond to three independent relational invariants of the Jordan algebra. The Gram matrix of K_ω has rank ≤ 3 because the Jordan algebra A_J supports exactly three independent symmetric bilinear invariants. The closure defect $\Xi_C(D) = (D-3)_+$ measures excess rank — non-tautological, and rank-3 is forced by the algebra's representation theory, not by counting channels.

Technique. Fubini-Study metric (quantum information geometry) + Jordan algebra representation theory

***Theorem to prove.** Theorem (D=3 from Channel-Rank): The Gram matrix of the phase-preserving kernel K_ω has rank ≤ 3 for generic admissible sets $\{A_i\} \subset A_{loc}$, with equality iff the three inference channels (existence, distance, direction) are independently supported by the Jordan algebra structure.*

Derivation sketch.

(1) $K_\omega(A,B) = \omega(A^\dagger B) / \sqrt{(\omega(A^\dagger A) \cdot \omega(B^\dagger B))}$ is the cosine of the angle between $[A]_\omega$ and $[B]_\omega$ in GNS Hilbert space — standard.

(2) $d_{FS} = \ell_0 \cdot \arccos(|K_\omega|)$ is the Fubini-Study geodesic distance on projective Hilbert space. Triangle inequality is automatic (Petz 1996, standard quantum information geometry).

(3) The three inference channels: Existence = $|K_\omega| > 0$ (Quadratic); Distance = d_{FS} (Cubic); Direction = $\arccos$ phase of K_ω (Cubic). These are the three independent invariants of the Jordan product $A \circ B = \frac{1}{2}(AB+BA)$.

(4) The Gram matrix $G_{ij} = K_\omega(A_i, A_j)$ for D points has rank = $\min(D, \text{rank of Jordan invariant space})$. If the Jordan algebra supports exactly 3 independent invariants, $\text{rank}(G) \leq 3$, with equality for generic admissible sets.

(5) $\Xi_C(D) = (D-3)_+ = \text{excess rank}$. $\Xi_C(D) = 0$ iff $D \leq 3$. This is non-tautological (rank is computed from the algebra, not defined as $|D-3|$).

Dependencies.

- Fubini-Study metric adoption (easy — see Ch 3 resolution)
- Jordan algebra representation theory argument: prove A_J supports exactly 3 independent symmetric bilinear invariants (HARD — this is a representation-theoretic question, not a calculation)
- The K_ω Jordan-built problem (F2 from patch plan): current $K_\omega = |\omega(A^\dagger B)|^2/\dots$ does NOT use $A \circ B$. Fix: $K_\omega^J(A, B) = \omega(A \circ B)/\sqrt{(\dots)}$. Fubini-Study still works with K_ω^J ($\arccos$ on any inner product).

Ripple effects. KILLS: Factorization (Condition 3.3.1), Dominance (Condition 3.3.2). REDEFINES: cubic kernel $V_\omega = |\det K_\omega|$ becomes the Riemannian volume element of the FS metric (standard differential geometry). UNAFFECTED: κ_B (uses ℓ_0 from algebra), BH entropy (uses ℓ_0 from spectral discreteness). IMPROVES: tangent space becomes genuine Riemannian; direction becomes real tangent vector. OPENS: the K_ω^J variant (Jordan-built kernel) — if adopted, the $D=3$ Gram-rank argument becomes a clean representation-theoretic result.

Disagreements resolved. One reviewer proposed $D=3$ from 'position = 2 channels (distance + direction).' This is the right intuition but the wrong mechanism — the triangle inequality doesn't come from counting channels, it comes from using a geodesic distance. The channel-counting gives $D=3$ via Gram rank; the metricity comes from Fubini-Study. These are separate fixes for separate problems, both needed.

Status. Strategy-level. Fubini-Study swap is easy; Jordan representation argument is hard. Together they close $D=3$ and the triangle inequality.

4.2 [T2] Relational Burden = Cross-Sector Burden (CLOSEST TO DONE)

► [T2] Relational Burden = Cross-Sector Burden

Priority: CLOSEST TO DONE | v2.0 location: Def 2.6.2 (B_{cross}), Def 3.9.1 (Π_{net}), Def 4.4.1 (three-channel burden), Thm 4.4.3 (relational burden = DM)

The problem. v2.0 defines relational burden as $\Theta^{\text{rel}} = [\hat{C}_\alpha, \Pi_{\text{net}}]$ — a commutator with the network operator. This is abstract and doesn't clearly connect to the physical intuition that relational burden lives 'in the gaps between sectors.' The dark-matter identification (Thm 4.4.3) is structurally motivated but the quantitative

profile (rotation curves, halo shapes) is quarantined as Q-1 OPEN.

The proposed approach. Decompose $F = F_{\text{intra}} + F_{\text{cross}}$ where $F_{\text{intra}} = \sum_i P_i F P_i$ (mode/existence burden, sources baryonic matter) and $F_{\text{cross}} = \sum_{\{i \neq j\}} P_i F P_j$ (relational burden, sources dark matter). Then: • $B_{\text{mode}} = \text{Tr}(\rho \cdot F_{\text{intra}})$ — localized inside sectors, sources baryonic mass ($m^2 \equiv B_0$) • $B_{\text{rel}} = \text{Tr}(\rho \cdot F_{\text{cross}}) = \sum_{\{i \neq j\}} B_{\text{cross}}(i,j)$ — lives in the gaps between sectors, sources dark matter The identification $\Theta^{\wedge}(\text{rel}) = [\hat{C}_{\alpha}, \Pi_{\text{net}}] = \sum_{\{i \neq j\}} K_{\omega}(i,j) \cdot [\hat{C}_{\alpha}, P_{ij}]$ is the network-weighted sum of cross-sector commutators. Relational burden IS cross-sector burden, evaluated at the network level.

Technique. Operator block decomposition ($F = F_{\text{intra}} + F_{\text{cross}}$) + graph Laplacian interpretation

***Theorem to prove.** Theorem (Relational Burden = Cross-Sector Linking Cost): $B_{\text{rel}} = \text{Tr}(\rho \cdot F_{\text{cross}}) = \sum_{\{i \neq j\}} K_{\omega}(i,j) \cdot B_{\text{cross}}(i,j)$, where $B_{\text{cross}}(i,j) = \sum_{\alpha} w_{\alpha} \cdot \omega([\hat{C}_{\alpha}, P_{ij}]^{\dagger} [\hat{C}_{\alpha}, P_{ij}])$. This is algebraic (not probabilistic), non-local (extends beyond visible matter), non-luminous (commutator structure, no EM coupling), and sources gravity via $\Theta^{\wedge}(B)_{\mu\nu}$.*

Derivation sketch.

- (1) $F = \sum_{\{\alpha, \beta\}} \pi(\Delta_{\alpha\beta})^{\dagger} \pi(\Delta_{\alpha\beta})$ where $\Delta_{\alpha\beta} = [\hat{C}_{\alpha}, \hat{C}_{\beta}]$. Decompose via sector projectors: $F = \sum_i P_i F P_i + \sum_{\{i \neq j\}} P_i F P_j = F_{\text{intra}} + F_{\text{cross}}$.
- (2) F_{intra} captures constraint violations WITHIN each sector (mode/existence burden — the cost of maintaining zero-preservation inside a single localized sub-algebra).
- (3) F_{cross} captures constraint violations ACROSS sectors (relational burden — the cost of maintaining coherence between sectors i and j via the cross-transition operator P_{ij}).
- (4) $B_{\text{rel}} = \text{Tr}(\rho \cdot F_{\text{cross}}) = \sum_{\{i \neq j\}} \sum_{\alpha} w_{\alpha} \cdot \omega([\hat{C}_{\alpha}, P_{ij}]^{\dagger} [\hat{C}_{\alpha}, P_{ij}]) = \sum_{\{i \neq j\}} K_{\omega}(i,j) \cdot B_{\text{cross}}(i,j)$. The $K_{\omega}(i,j)$ weighting provides natural cutoff (weakly-linked sectors contribute little).
- (5) The (sector count)² scaling concern: summing over all pairs scales as N^2 . Resolution: $K_{\omega}(i,j)$ decays with distance, so effectively only $O(N)$ pairs contribute significantly — matching smooth-halo phenomenology.

Dependencies.

- The fracture mechanism must be ONE mechanism (not two — currently record-sector decoherence and extension-sector fracture are separately defined). The unified sector set is needed.
- P_{ij} (cross-transition operator) must be explicitly defined (partially done in Def 2.6.2, needs completion).
- The $K_{\omega}(i,j)$ weighting's decay properties need verification (does it decay fast enough to prevent N^2 blowup?).

Ripple effects. IMPROVES: dark matter profile becomes computable (Q-1 moves from OPEN to STRUCTURAL — mechanism established, amplitudes require simulation). UNAFFECTED: FIREWALL (F_{cross} is algebraic, doesn't cross the probabilistic/algebraic divide). CLEANER: mass-burden identity $m^2 \equiv B_0$ now uses B_{intra} (cleaner separation between baryonic and dark sectors). OPENS: the 'cosmic web' interpretation — relational burden naturally forms a network of interfacial tension that mimics dark-matter halo structure.

Disagreements resolved. Multiple voices proposed relational burden as 'cross-sector burden' but with different formulas. The resolution: $B_{\text{rel}} = \text{Tr}(\rho \cdot F_{\text{cross}})$ is the cleanest because it comes from the operator decomposition (no new primitives). The alternative formula $B_{\text{rel}} = \sum K_{\omega(i,j)} \cdot B_{\text{cross}(i,j)}$ is equivalent (it's the same sum, just expanded). The key insight all voices agree on: relational burden is NOT a volumetric cost — it's an INTERFACIAL cost (the tension between sectors).

Status. Strategy-level. The decomposition is definitional (easy). The halo-profile computation requires numerical simulation (medium). The unified-sector-set prerequisite is conceptual (needs the fracture mechanism to be one mechanism, not two).

4.3 [T3] T-2 (Spectral Gap) (STRATEGY OUTLINE)

► [T3] T-2 (Spectral Gap)

Priority: STRATEGY OUTLINE | v2.0 location: Thm 1.6.1 (Convergence Theorem), T-2 (§7.2 Table 7.2.1)

The problem. T-2 asks: does F have a spectral gap on $\ker(M)$ as a Hilbert space (infinite-dimensional)? v2.0 verifies the gap only in a finite-dim toy model ($M_6(\mathbb{C})$). The infinite-dimensional generalization is the existential open problem — if T-2 fails, the Convergence Theorem, thin=full, mass-burden, Dirac bracket, $\Lambda_B=0$, Einstein closure, Friedmann, and BH holography all collapse from Established to Conditional.

The proposed approach. Several voices propose 'T-2 via fixed $D=3$ ' — the idea that $D=3$ being fixed means the Hilbert space is finite-dimensional. This is WRONG (one reviewer correctly flags it): $D=3$ fixes the SPATIAL dimension, not the HILBERT-SPACE dimension. Ordinary 3D quantum mechanics has an infinite-dimensional Hilbert space. The correct path: F is a FRUSTRATION-FREE HAMILTONIAN ($F = \sum_{\{\alpha,\beta\}} f_{\alpha\beta}$ where $f_{\alpha\beta} = \pi(\Delta_{\alpha\beta})\dagger\pi(\Delta_{\alpha\beta}) \geq 0$; $\ker(F) = \cap_{\{\alpha,\beta\}} \ker(f_{\alpha\beta}) = \ker(M)$). Use the DETECTABILITY LEMMA (Aharonov-Arad-Landau-Vazirani 2009) or KNABE'S BOUND (1988) to lift LOCAL spectral gaps to a GLOBAL gap. The key result: if the local gap γ_{local} is UNIFORM (doesn't shrink as the universe expands), the global gap is bounded below by $\gamma_{\text{local}}/(C \cdot \Delta)$ where Δ is the max interaction degree (bounded because $D=3$ is fixed $\rightarrow$ each cell has ≤ 6 neighbors). REFORMULATION: T-2' = uniform local gap + Lieb-Robinson decay $\Rightarrow$ global convergence. This is standard many-body theory. The 'infinite-dimensional' aspect becomes a thermodynamic limit (infinitely many finite-dimensional sectors, not infinitely large sectors).

Technique. Frustration-free Hamiltonians + detectability lemma (Aharonov et al. 2009) + Knabe's bound (1988)

***Theorem to prove.** Theorem (T-2': Uniform Local Gap $\Rightarrow$ Global Convergence): If (i) each local sector H_k has a spectral gap $\gamma_{\text{local}} > 0$ that is uniform across sectors, and (ii) the constraint index set $\{\hat{C}_{\alpha}\}$ has a well-defined adjacency structure (from Jordan co-occurrence) with bounded degree Δ , then the global Reconciliation Propagator R_t has a spectral gap $\gamma_{\text{global}} \geq \gamma_{\text{local}}/(C \cdot \Delta) > 0$, and the Convergence Theorem holds in the thermodynamic limit.*

Derivation sketch.

(1) F is frustration-free: $F = \sum f_{\alpha\beta}$ with $f_{\alpha\beta} \geq 0$ and $\ker(F) = \cap \ker(f_{\alpha\beta})$. This is the standard frustration-free structure (Nachtergaele 1996).

(2) Each local sector H_k is finite-dimensional (bounded by the ℓ_0 -floor: each sector has $\leq N_{\text{cell}} \sim \text{vol}/\ell_0^3$ reconciliation cells). On finite-dim H_k , the local gap $\gamma_{\text{local}} > 0$ is guaranteed by LaSalle (standard Lyapunov theory for gradient descent on a Morse function).

(3) The detectability lemma gives: $\gamma_{\text{global}} \geq \gamma_{\text{local}} / (C \cdot \Delta)$ where $\Delta = \max$ interaction degree (≤ 6 in 3D) and C depends on commutation structure. Since $D=3$ is fixed, Δ is bounded.

(4) The UNIFORMITY condition (γ_{local} doesn't shrink as new cells are added) is the key remaining step. Plausible because the local algebra per cell is always the same $M_d(\mathbb{C})$ — so the local gap is always the same. Needs proof.

(5) Cross-sector decay: Lieb-Robinson bounds (see T-1) give exponential decay of correlations outside the causal cone. This ensures the thermodynamic limit ($N \rightarrow \infty$ sectors) preserves convergence.

Dependencies.

- **ADJACENCY STRUCTURE** on $\{\hat{C}_\alpha\}$: currently there is NO notion of 'local' for the constraint index set. This is the single biggest missing ingredient. Proposed source: Jordan co-occurrence ($A \circ B$) defines 'who's near whom' — two constraints are adjacent if their Jordan product is non-vanishing.
- **LOCAL GAP VERIFICATION**: need to verify $\gamma_{\text{local}} > 0$ on the specific constraint algebra of RCF (not just on the $M_6(\mathbb{C})$ toy model).
- **UNIFORMITY PROOF**: need to show γ_{local} doesn't shrink under Open Expansion (the local algebra is always $M_d(\mathbb{C})$, but the proof requires checking that the constraint structure is translation-invariant in the emergent geometry).

Ripple effects. CLOSES (if successful): Convergence Theorem (Thm 1.6.1), thin=full (Thm 1.7.4), mass-burden identity (Thm 2.5.1), Dirac bracket (Thm 1.4.6), $\Lambda_B=0$ (Cor 4.5.5), Einstein closure (Thm 4.5.4). UNIFIES WITH T-1: the spectral gap of F determines both the convergence rate (T-2) and the causal speed $c = \gamma \cdot \ell_0$ (T-1) — see Ch 7. OPENS: the adjacency structure on $\{\hat{C}_\alpha\}$ (from Jordan co-occurrence) is the same ingredient T-1 needs for Lieb-Robinson.

Disagreements resolved. RESOLVED: 'T-2 via fixed $D=3$ ' is WRONG. $D=3$ fixes spatial dimension, not Hilbert-space dimension. The correct mechanism is: $D=3$ fixes the LOCAL ALGEBRA ($M_d(\mathbb{C})$), which gives finite-dimensional local sectors, and the detectability lemma lifts local gaps to global. The 'dimensions don't scale' intuition is right but the mechanism is different from what several voices claimed.

Status. Strategy outline. The detectability lemma is established mathematics. The missing ingredient is the adjacency structure on $\{\hat{C}_\alpha\}$ (from Jordan co-occurrence) — this is the same prerequisite T-1 needs. Once the adjacency exists, T-2 reduces to verifying the local gap and proving uniformity.

4.4 [T4] T-1 (γ Derivation) (STRATEGY OUTLINE)

► [T4] T-1 (γ Derivation)

Priority: STRATEGY OUTLINE | v2.0 location: Thm 1.1.2 ($c = \gamma \cdot \ell_0$), T-1 (§7.2 Table 7.2.1), Post 1.1.1 (Open Expansion Principle)

The problem. T-1 asks: derive γ (the Open Expansion rate) from algebraic primitives, so that $c = \gamma \cdot \ell_0$ is a derived constant rather than a postulated bound. v2.0's Thm 1.1.2 is labeled 'Conditional' but the proof sketch is a dimensional argument ('each incremental expansion requires time at least $\varepsilon = 1/\gamma$ ') — closer to a heuristic scaling relation than a derived bound. The patch plan (M7) recommends downgrading to 'Heuristic Scaling Relation' until T-1 closes.

The proposed approach. Map to LIEB-ROBINSON BOUNDS (established 1972, extended many times). For systems with LOCAL interactions, $[A(t), B]$ stays exponentially small until t exceeds $d(A, B)/v_{LR}$, and v_{LR} is DERIVED from the interaction structure — not postulated. In RCF: the Open Expansion generator is $H_{eff} = \sqrt{F + \delta F}$. Its Lieb-Robinson velocity IS $c = \gamma \cdot \ell_0$. The proposal: $\gamma := \sqrt{\lambda_{min}(F)}$ where $\lambda_{min}(F)$ is the smallest non-zero eigenvalue of F (the spectral gap on the orthogonal complement of $\ker(F)$). This UNIFIES T-1 and T-2: the same spectral gap determines both the convergence rate (T-2) and the causal speed (T-1). The speed limit $c = \ell_0 \cdot \sqrt{\lambda_{min}(F)}$ is a derived constant of the algebra.

Technique. Lieb-Robinson bounds (1972) + spectral gap of F (shared with T-2)

***Theorem to prove.** Theorem (T-1': γ from F Spectral Gap): If the constraint index set $\{\hat{C}_\alpha\}$ has a well-defined adjacency structure (from Jordan co-occurrence) with bounded degree, then the Open Expansion generator $H_{eff} = \sqrt{F + \delta F}$ obeys a Lieb-Robinson bound with velocity $v_{LR} = \ell_0 \cdot \sqrt{\lambda_{min}(F)}$, where $\lambda_{min}(F)$ is the spectral gap of F on the orthogonal complement of $\ker(F)$. The causal speed limit is $c = v_{LR} = \ell_0 \cdot \sqrt{\lambda_{min}(F)}$.*

Derivation sketch.

(1) $F = \sum_{\alpha, \beta} \pi(\Delta_{\alpha\beta})^\dagger \pi(\Delta_{\alpha\beta})$ where $\Delta_{\alpha\beta} = [\hat{C}_\alpha, \hat{C}_\beta]$. Each $\Delta_{\alpha\beta}$ is a local Lie bracket of constraints. If constraints are ultralocal (supported on regions $\sim \ell_0$), then $\Delta_{\alpha\beta}$ is supported on the union of two ℓ_0 -regions.

(2) The Lieb-Robinson velocity for $H_{eff} = \sqrt{F}$ is: $v_{LR} \sim \|\sqrt{F}\|_{local} \sim \sqrt{(\sum_{\alpha, \beta} \|\Delta_{\alpha\beta}\|^2)}$.

(3) On the physical sector, $F \rightarrow 0$. Near the physical sector, the relevant scale is the lowest non-zero eigenvalue $\lambda_{min}(F)$ — the spectral gap on the orthogonal complement of $\ker(F)$.

(4) Propose: $\gamma := \sqrt{\lambda_{min}(F)}$. Then $c = \ell_0 \cdot \gamma = \ell_0 \cdot \sqrt{\lambda_{min}(F)}$. This is derived from the spectral structure of F , not postulated.

(5) The Lorentz factor: a moving sector has additional interaction burden $B_{int}(v) \propto v^2$ (leading-order cross-sector commutator expansion). Matching to $\sqrt{(1-v^2/c^2)}$: $\lambda \cdot B_{int}(v) = v^2/(2c^2)$, giving $B_{int}(v) = v^2/(2\lambda \cdot \gamma^2 \cdot \ell_0^2)$. The Lorentz factor $\gamma_L = 1/\sqrt{(1-v^2/c^2)}$ emerges from the burden cost of relative motion.

Dependencies.

- ADJACENCY STRUCTURE on $\{\hat{C}_\alpha\}$ (same as T-2 — from Jordan co-occurrence). This is the single shared prerequisite.
- SPECTRAL GAP of F (same as T-2 — $\lambda_{min}(F) > 0$). T-1 and T-2 are unified: the same gap determines both.
- EXACT $B_{int}(v)$ derivation: need to derive v^2 scaling from the cross-sector commutator $[\hat{C}_\alpha, P_{kj}]$ evaluated on a moving sector state. The square root (rather than linear) in the Lorentz factor needs derivation,

not definition.

Ripple effects. CLOSES: T-1 (γ derived, $c = \ell_0 \cdot \sqrt{\lambda_{\min}(F)}$). UNIFIES: T-1 and T-2 both reduce to the spectral gap of F — one mathematical object, two physical consequences. OPENS: the Lorentz factor derivation path (from burden cost of relative motion). DOWNGRADES: Thm 1.1.2 from 'Heuristic Scaling Relation' back to 'Theorem (conditional on T-2)' once the Lieb-Robinson bound is established.

Disagreements resolved. RESOLVED: $\gamma = \sqrt{\lambda_{\min}(F)}$ needs care because $F \rightarrow 0$ on the physical sector. The resolution: $\lambda_{\min}$ is the gap on the ORTHOGONAL COMPLEMENT of $\ker(F)$, i.e., the smallest non-zero eigenvalue. This is the same gap T-2 needs — confirming the unification.

Status. Strategy outline. The Lieb-Robinson technique is established. The missing ingredient is the adjacency structure on $\{\hat{C}_\alpha\}$ (same as T-2). Once adjacency exists, T-1 reduces to computing $\lambda_{\min}(F)$ and applying the standard Lieb-Robinson bound. The Lorentz factor derivation is harder (needs the exact $B_{\text{int}}(v)$ form).

4.5 [T5] T-4 / Z-Envariance (Born Rule) (STATUS ASSESSMENT)

► [T5] T-4 / Z-Envariance (Born Rule)

Priority: STATUS ASSESSMENT | v2.0 location: Def 2.7.3 (Z-envariance), Thm 2.7.4 (Born rule), T-4 (§7.2)

The problem. v2.0 claims T-4 is CLOSED and the Born rule is a theorem. But the patch plan (F4) flagged that Z-envariance (Def 2.7.3) is T-2-conditional, so T-4 is 'STRENGTHENED (conditional on T-2),' not 'CLOSED.' The brainstorming proposes 'least resistance' — minimize burden to get $p_i = |c_i|^2$ — but one reviewer correctly flags a CIRCULARITY RISK: the burden cost functional already encodes $|c_i|^2$ via the density matrix structure.

The proposed approach. Two directions, one more promising: (A) 'LEAST RESISTANCE' — CIRCULARITY RISK. The proposed burden cost $B_{\text{decomp}}(\{p_k\}) = \sum p_k \cdot B_{\text{intra}}(k) + \sum \sqrt{p_k \cdot p_j} \cdot B_{\text{cross}}(k, j)$ already contains $\sqrt{p_k \cdot p_j}$ terms — which come from the off-diagonal matrix elements of $\rho = \sum p_k |k\rangle\langle k| + \text{cross terms}$. So the $|c_i|^2$ structure is BUILT INTO the burden functional via the density matrix. 'Minimizing burden yields $p_i = |c_i|^2$ ' may be tautological unless B_{cross} is defined INDEPENDENTLY of the sector weights p_k . This direction needs work to escape circularity. (B) PETZ/BURES METRIC — MORE PROMISING. The Open Extension flow is ALREADY gradient descent on the Bures metric (§1.5.3). Petz's 1996 classification theorem shows the Bures metric is singled out by extra symmetry requirements among monotone quantum metrics. It's tightly linked to the Fubini-Study metric on pure states (which encodes $|c_i|^2$ naturally — the Fubini-Study distance between $|\psi\rangle = \sum c_i |i\rangle$ and $|i\rangle$ is $\arccos(|c_i|)$, so the Born weight is geometrically privileged). If p_i is 'how much Bures-weight sector i carries,' the Born rule falls out of the METRIC the dynamics already runs on — not from a separate minimization. This is a real direction, not yet built. (C) Z-ENVARIANCE AS LABEL-PERMUTATION SYMMETRY — FREE. F is built from compatibility brackets $\Delta_{\alpha\beta} = [\hat{C}_\alpha, \hat{C}_\beta]$ that don't know about sector labels a priori. So the burden landscape IS permutation-symmetric at the fixed point. This gives phase invariance + equal-magnitude equality for free. Combined with additivity (from M_{red} 's sectoral decomposition, Thm 2.7.1), the standard functional equation yields $p_i = |c_i|^2$.

Technique. Petz monotone metric classification (1996) + Fubini-Study geometry + label-permutation symmetry

Theorem to prove. *Theorem (Born from Bures Geometry, CONDITIONAL): If (i) the Open Extension flow is gradient descent on the Bures metric (Established, §1.5.3), (ii) the Bures metric is the unique monotone quantum metric satisfying Petz's symmetry requirements (Established, Petz 1996), and (iii) sector weights p_i measure the Bures-weight of each sector in the desynchronized composite (Thm 1.6.3), then $p_i = |c_i|^2$ is the unique consistent assignment (Born rule). CONDITIONAL on T-2 (for the fixed-point existence).*

Derivation sketch.

(1) Open Extension = gradient descent on Bures metric (§1.5.3, Established). The Bures metric for pure states reduces to the Fubini-Study metric.

(2) Petz 1996: the Bures metric is the unique monotone quantum metric satisfying (a) unitary invariance, (b) joint convexity, (c) the data-processing inequality. Standard quantum information geometry.

(3) Fubini-Study distance: $d_{FS}(|\psi\rangle, |i\rangle) = \arccos(|\langle i|\psi\rangle|) = \arccos(|c_i|)$. The Born weight $|c_i|^2 = \cos^2(d_{FS})$ is the geometrically privileged measure of 'how close $|\psi\rangle$ is to $|i\rangle$.'

(4) Label-permutation symmetry: $F = \sum \Delta_{\alpha\beta} \dagger \Delta_{\alpha\beta}$ is invariant under sector relabeling (the sum doesn't depend on labels). So the fixed-point burden landscape is permutation-symmetric — giving phase invariance + equal-magnitude equality for free.

(5) Additivity from M_{red} : Thm 2.7.1 (sectorwise zero-decomposition, now with genuine content via M_{red}) gives the additivity premise. Phase invariance + equal-magnitude equality + additivity $\Rightarrow p_i = |c_i|^2$ via the standard functional equation (Thm 2.7.4).

Dependencies.

- T-2 (for fixed-point existence — the desynchronized composite must exist for the Born rule to be well-defined). This is why T-4 is CONDITIONAL on T-2, not CLOSED.
- Fubini-Study metric adoption (Ch 3 resolution — connects the Born rule to the same geometry as $D=3$ and the triangle inequality).
- M_{red} 's sectoral structure (Already defined at §1.4.3 — gives the additivity premise genuine content).

Ripple effects. DOWNGRADES: T-4 from 'CLOSED' to 'STRENGTHENED (conditional on T-2)' (consistent with patch plan F4). IMPROVES: the Born rule's objectivity premise now has genuine content (from M_{red} 's sectoral structure + Bures geometry, not from a vacuous Thm 7.4). CONNECTS: the Born rule to the same Fubini-Study geometry that fixes $D=3$ and the triangle inequality — a pleasing unification.

Disagreements resolved. RESOLVED: the 'least resistance' direction (A) has a circularity risk. The Petz/Bures direction (B) is more promising because it derives the Born rule from the metric the dynamics ALREADY uses, not from a separate minimization. The label-permutation symmetry (C) is free and gives two of the three Born-rule premises. The third premise (additivity) comes from M_{red} . NET: T-4 is STRENGTHENED but not CLOSED — conditional on T-2.

Status. Status assessment. The Petz/Bures direction is the most promising but needs development. The 'least resistance' direction has a circularity risk that must be addressed (define B_{cross} independently of p_k). T-4 remains STRENGTHENED (conditional on T-2), not CLOSED.

4.6 [T6] T-6 (Entropy-Area Scaling) (STATUS ASSESSMENT)

► [T6] T-6 (Entropy-Area Scaling)

Priority: STATUS ASSESSMENT | v2.0 location: Thm 5.3.1 (dimensional suppression), Thm 5.5.3 (entropy-area), T-6 (§7.2)

The problem. v2.0 derives area scaling $S \propto \text{Area}(\partial R)$ but honestly flags that the $1/4$ coefficient is NOT derived — the proof assumes exponential microstate growth (the conclusion). T-6 is 'PARTIAL: area-scaling established, coefficient open.' The brainstorming proposes channel suppression as the mechanism: at black-hole saturation, the radial inference channel collapses, leaving rank-2 boundary channels.

The proposed approach. The channel-suppression mechanism: at black-hole saturation ($\Pi_{ij} \rightarrow \Pi_{\max}$), the radial inference channel $\Delta^{\text{rad}} \rightarrow 0$. Remaining boundary channels: existence + 2 tangential = rank 2. Entropy = log of boundary microstate count; with rank-2 channels on a 2D surface, $N \sim \exp(\alpha \cdot \text{Area}/\ell_0^2)$, giving $S = \alpha \cdot \text{Area}/\ell_0^2$. The area scaling is structurally derived (rank drops from 3 to 2 at the boundary $\rightarrow$ 2D channel structure $\rightarrow$ area scaling). The coefficient $\alpha = 1/4$ requires showing each ℓ_0^2 boundary cell carries exactly $\ln 2$ bits (existence binary, tangential directions frozen at saturation). The HARDER question (one reviewer's upgrade): if '3D $\rightarrow$ 2D at horizon' means rank dropping from 3 to 2, counting microstates becomes an actual GEOMETRIC problem — how many independent rank-2 configurations exist — rather than an assumed ansatz. This is harder than D=3, not solved, but a real question now.

Technique. Inference-channel rank collapse + boundary microstate counting

***Theorem to prove.** Theorem (Entropy-Area from Channel Suppression, CONDITIONAL on D=3): If (i) D=3 is established (Ch 4, three inference channels), (ii) at black-hole saturation the radial channel collapses ($\Delta^{\text{rad}} \rightarrow 0$, Thm 5.3.1), and (iii) each ℓ_0^2 boundary cell carries exactly $\ln 2$ bits (existence binary, tangential directions frozen), then $S_{\partial R} = \text{Area}(\partial R)/(4\ell_0^2) = \text{Area}(\partial R)/(4\ell_P^2)$ after the $\ell_0 \leftrightarrow \ell_P$ identification.*

Derivation sketch.

- (1) Channel rank in bulk: $r = 3$ (existence + distance + direction).
- (2) Channel rank at boundary: radial direction collapses ($\Delta^{\text{rad}} \rightarrow 0$ at ∂R , Thm 5.3.1), leaving $r = 2$ (existence + 2 tangential).
- (3) Boundary microstate count: partition ∂R into cells of area $\sim \ell_0^2$. Each cell supports existence (binary: present/absent $\rightarrow \ln 2$ nats) and 2 tangential directions (continuous angles, regulated by ℓ_0 discreteness).
- (4) At saturation ($\Pi_{ij} \rightarrow \Pi_{\max}$), tangential directions freeze (only binary existence remains active per cell). With 1 bit per cell: $N_{\partial R} \sim 2^{(\text{Area}/\ell_0^2)}$, so $S = \ln N = (\ln 2) \cdot \text{Area}/\ell_0^2$.
- (5) Identifying $\ell_0^2 = 4\ell_P^2 \cdot \ln 2$ (from κ_B calibration via Newtonian limit, Thm 4.6.1) gives $S = \text{Area}/(4\ell_P^2)$. NOT circular — it's a consistency condition between three independent derivations (κ_B , $S_{\partial R}$, c).

Dependencies.

- D=3 closure (Ch 4 — the three-channel formalism must be established first).

- The $\ell_0 \leftrightarrow \ell_P$ identification (from κ_B calibration, Thm 4.6.1 — already established as 'calibrated by Newtonian limit').
- The freezing of tangential directions at saturation ($\Pi_{ij} \rightarrow \Pi_{\max}$ must freeze ALL directional fluctuations, not just radial — needs verification).

Ripple effects. CLOSES (if successful): T-6 (BH entropy coefficient). UPGRADES: Q-8 (holography) from PARTIAL to STRUCTURAL (the 2D boundary is derived from channel collapse, not assumed). CONNECTS: the coefficient 1/4 becomes a consistency condition between κ_B (Newtonian limit), $S_{\partial R}$ (channel counting), and c (spectral gap) — three independent derivations that must agree.

Disagreements resolved. RESOLVED: the v2.0 proof was circular (assumed exponential microstate growth = the conclusion). The channel-suppression approach is non-circular IF the cell-counting argument works: it counts actual geometric configurations (rank-2 boundary states), not an assumed exponential ansatz. The coefficient 1/4 is NOT derived by the channel argument alone — it requires the $\ell_0 \leftrightarrow \ell_P$ calibration from κ_B . This is a consistency condition, not circularity.

Status. Status assessment. Area scaling is STRUCTURALLY DERIVED (conditional on $D=3$). The coefficient 1/4 requires the cell-counting argument (existence binary + tangential frozen at saturation) PLUS the $\ell_0 \leftrightarrow \ell_P$ calibration from κ_B . Harder than $D=3$, not solved, but a real question now instead of a placeholder.

Chapter 5 — Ripple-Effect Trace

For each proposed change, what changes downstream. The five changes with significant ripple effects:

ID	Change	Ripple Effects
R1	Fubini-Study metric adoption ($d_{\omega} = l_0 * \arccos(\sqrt{K_{\omega}})$)	KILLS: Factorization (3.3.1), Dominance (3.3.2). REDEFINES: cubic kernel $V_{\omega} = \det K_{\tilde{\omega}} $ becomes Riemannian volume element of FS metric. UNAFFECTED: κ_B (uses l_0 from algebra), BH entropy (uses l_0 from spectral discreteness). IMPROVES: triangle inequality automatic; tangent space becomes genuine Riemannian.
R2	$F_{\hat{}} = F_{\text{intra}} + F_{\text{cross}}$ decomposition	IMPROVES: relational burden = $\text{Tr}(\rho * F_{\text{cross}})$; dark matter profile computable (Q-1 -> STRUCTURAL). UNAFFECTED: FIREWALL (F_{cross} is algebraic). CLEANER: mass-burden $m^2 = B_0$ uses B_{intra} (cleaner baryonic/dark separation). OPENS: cosmic-web interpretation of dark matter halos.
R3	$\gamma = \sqrt{\lambda_{\min}(F_{\hat{}})}$	CLOSES: T-1 (γ derived, $c = l_0 * \sqrt{\lambda_{\min}(F_{\hat{}})}$). UNIFIES: T-1 and T-2 both reduce to spectral gap of $F_{\hat{}}$. OPENS: Lorentz factor derivation path (from burden cost of relative motion). DOWNGRADES: Thm 1.1.2 from 'Heuristic' back to 'Theorem (conditional on T-2)'.
R4	K_{ω}^J (Jordan-built kernel: $K_{\omega}^J(A,B) = \omega(A \text{ circ } B) / \sqrt{\dots}$)	FIXES: F2 from patch plan (K_{ω} now genuinely uses Jordan product). UNAFFECTED: Fubini-Study still works ($\arccos$ on any inner product). CHANGES: D=3 Gram-rank argument becomes representation-theoretic (about Jordan algebra invariants, not channel counting). RIPPLE: cubic kernel now uses K_{ω}^J .
R5	Adjacency structure on $\{C_{\alpha}\}$ from Jordan co-occurrence	ENABLES: Lieb-Robinson bounds (T-1). ENABLES: detectability lemma (T-2). UNIFIES: the two-primitive picture (Jordan = locality graph, Lie = dynamics on graph). PREREQUISITE for: T-1, T-2, and the Lorentz factor derivation.

The Key Ripple Insight

The Fubini-Study swap looks scary because $-\log(K_{\omega})$ appears in κ_B and BH entropy. But the trace shows that l_0 in those formulas comes from the ALGEBRA (spectral discreteness of $F_{\hat{}}$, the l_0 -floor), not from the distance function. Switching the distance from $-l_0 * \log(K)$ to $l_0 * \arccos(\sqrt{K})$ keeps the same l_0 scale (the algebra doesn't change) — only the SHAPE of the distance changes. κ_B and BH entropy depend on l_0 as a SCALE, not on the shape of d_{ω} , so they survive unchanged. See Lemma L6 in the deep-dive appendix.

Chapter 6 — Deep-Dive Appendix: Fubini-Study Resolution to Lemma-Skeleton Level

This appendix develops the Fubini-Study metric resolution (Chapter 3) to lemma-skeleton level: formal lemma statements with 2–3 sentence proof sketches and explicit dependency chains. Ready to be expanded into full proofs in Phase C.

L1 — Lemma (Fubini-Study is a metric). The function $d_{\text{FS}}: \mathbb{P}(\mathcal{H}_{\text{kin}}) \times \mathbb{P}(\mathcal{H}_{\text{kin}}) \rightarrow [0, \pi]$...

Statement. Lemma (Fubini-Study is a metric). The function $d_{\text{FS}}: \mathbb{P}(\mathcal{H}_{\text{kin}}) \times \mathbb{P}(\mathcal{H}_{\text{kin}}) \rightarrow [0, \pi/2]$ defined by $d_{\text{FS}}([A], [B]) = \arccos(|K_{\omega}(A, B)|) = \arccos(|\langle A|B \rangle_{\omega}| / (\|A\|_{\omega} \cdot \|B\|_{\omega}))$ is a metric on projective GNS space $\mathbb{P}(\mathcal{H}_{\text{kin}})$.

Proof sketch. Standard. d_{FS} is the geodesic distance on the projective Hilbert space $\mathbb{P}(\mathcal{H}_{\text{kin}})$ equipped with the Fubini-Study metric tensor. Non-negativity, symmetry, and identity of indiscernibles are immediate. The triangle inequality follows from the geodesic structure: $\arccos(|\langle \psi|\varphi \rangle|)$ is the length of the shortest geodesic between $[\psi]$ and $[\varphi]$ on $\mathbb{P}(\mathcal{H}_{\text{kin}})$, and geodesic distances satisfy the triangle inequality by definition.

References. Bengtsson & Życzkowski, *Geometry of Quantum States* (2006), §4.3. Petz 1996.

Dependencies. None (standard result).

L2 — Lemma (Triangle inequality is automatic). For any three admissible observables $A, B, C \in \mathcal{A}_{\text{loc}}$...

Statement. Lemma (Triangle inequality is automatic). For any three admissible observables $A, B, C \in \mathcal{A}_{\text{loc}}$, the Fubini-Study distance satisfies $d_{\text{FS}}(A, C) \leq d_{\text{FS}}(A, B) + d_{\text{FS}}(B, C)$ without any Factorization or Dominance condition.

Proof sketch. Direct corollary of L1. The geodesic distance on any Riemannian manifold satisfies the triangle inequality. $\mathbb{P}(\mathcal{H}_{\text{kin}})$ with the Fubini-Study metric is a Riemannian manifold. Therefore $d_{\text{FS}}(A, C) \leq d_{\text{FS}}(A, B) + d_{\text{FS}}(B, C)$ for all A, B, C — no conditions on the triad needed.

References. Follows from L1.

Dependencies. L1.

L3 — Lemma (Factorization and Dominance become unnecessary). Under the Fubini-Study d_{FS} ...

Statement. Lemma (Factorization and Dominance become unnecessary). Under the Fubini-Study distance, Condition 3.3.1 (Cubic Factorization) and Condition 3.3.2 (Cubic Dominance) are no longer required for the triangle inequality. They may be retained as properties of the kernel K_{ω} (not the distance) where needed downstream.

Proof sketch. By L2, the triangle inequality holds unconditionally. v2.0's Conditions 3.3.1 and 3.3.2 were introduced solely to force the triangle inequality for $d_{\omega} = -\ell_0 \cdot \log(K_{\omega})$. Under d_{FS} , these conditions are superfluous for metricity. The product structure $K_{\omega}(A, C) = K_{\omega}(A, B) \cdot K_{\omega}(B, C)$ (Factorization) remains a

property of the KERNEL K_ω (not the distance) and may still be used in the cubic kernel definition and downstream formulas.

References. Follows from L2.

Dependencies. L2.

L4 — Lemma (Log potential and Fubini-Study metric tensor). The function $\Phi(A,B) := -\log \dots$

Statement. Lemma (Log potential and Fubini-Study metric tensor). The function $\Phi(A,B) := -\log(K_\omega(A,B)) = -\log(|K_\omega(A,B)|^2) = -2 \cdot \log(|K_\omega(A,B)|)$ is a potential function whose Hessian gives the Fubini-Study metric tensor: $g^{FS}_{ij} = \frac{1}{2} \cdot \partial_i \partial_j \Phi|_{\text{diagonal}}$. The log is NOT a competitor to arccos — it is the potential whose Hessian IS the metric.

Proof sketch. Standard quantum information geometry. The Fubini-Study metric tensor on $\mathbb{P}(\mathcal{H}_{\text{kin}})$ in local coordinates $\{\theta^i\}$ is $g^{FS}_{ij} = \frac{1}{2} \cdot \partial^2(-\log|\langle \psi|\psi \rangle|^2)/\partial \theta^i \partial \theta^j$ evaluated at the diagonal. Equivalently, g^{FS} is the Hessian of the Rényi-2 relative entropy potential $-\log(K_\omega)$. This is the standard relationship between a Riemannian metric and its potential function (analogous to $g_{ij} = \partial_i \partial_j \varphi$ in Hessian geometry).

References. Petz 1996 (monotone metrics). Bengtsson & Życzkowski §4.4.

Dependencies. L1.

L5 — Lemma (Cubic kernel as Riemannian volume element). The corrected cubic volume el...

Statement. Lemma (Cubic kernel as Riemannian volume element). The corrected cubic volume element $V_\omega(A,B,C) = |\det[K_\omega(X_i, X_j)]|$ (Def 3.6.2, using phase-preserving kernel) is the Riemannian volume element of the Fubini-Study metric on the 3-dimensional submanifold spanned by $[A], [B], [C] \in \mathbb{P}(\mathcal{H}_{\text{kin}})$.

Proof sketch. The Gram determinant $|\det[K_\omega(X_i, X_j)]|$ is the squared volume of the parallelepiped spanned by $[X_1], [X_2], [X_3]$ in $\mathcal{H}_{\text{kin}}$. Under the Fubini-Study metric, the Riemannian volume element on the projective submanifold spanned by these three directions is precisely $\sqrt{(\det[g^{FS}_{ij}])} = |\det[K_\omega]|^{1/2}$ (up to a constant from the projective normalization). This is the standard relationship between Gram determinants and Riemannian volumes.

References. Standard differential geometry (Gram determinant = volume²).

Dependencies. L1, L4.

L6 — Lemma (κ_B and BH entropy inherit ℓ_0 from the algebra, not the distan...

Statement. Lemma (κ_B and BH entropy inherit ℓ_0 from the algebra, not the distance function). The formulas $\kappa_B = C/(\Pi_{\text{max}} \cdot \ell_0^2)$ and $S_{\partial R} = \text{Area}(\partial R)/(4\ell_0^2)$ use ℓ_0 as the fundamental scale from the ALGEBRA's spectral discreteness (the ℓ_0 -floor), not from the distance function. Therefore, switching d_ω from $-\ell_0 \cdot \log(K_\omega)$ to $d_{FS} = \ell_0 \cdot \arccos(\sqrt{K_\omega})$ does NOT change κ_B or $S_{\partial R}$.

Proof sketch. ℓ_0 appears in the framework as the fundamental relational scale from the spectral discreteness of F (the ℓ_0 -floor, Def 5.1.3: $\Pi_{\text{max}} = C \cdot \Pi / \ell_0^2$). The distance function $d_\omega = -\ell_0 \cdot \log(K_\omega)$ happens to use ℓ_0 as a prefactor, but ℓ_0 itself is defined independently (from the algebra, not from the distance). Switching to

$d_{FS} = \ell_0 \cdot \arccos(\sqrt{K_\omega})$ keeps the same ℓ_0 prefactor (the scale doesn't change) — only the SHAPE of the distance function changes ($\log \rightarrow \arccos$). The κ_B and BH formulas depend on ℓ_0 as a SCALE, not on the shape of d_ω , so they are unaffected.

References. Def 5.1.3 ($\Pi_{max} = C_\Pi/\ell_0^2$). Thm 4.5.6 (κ_B). Thm 5.5.3 ($S_{\partial R}$).

Dependencies. None (ℓ_0 is algebraic, not metric-dependent).

L7 — Lemma (K_ω^J Jordan-built variant). If K_ω is redefined as $K_\omega^J(A,B) := \omega(A \circ B) / \sqrt{(\omega(A^\dagger A) \cdot \omega(B^\dagger B))}$ where $A \circ B = \frac{1}{2}(AB + BA)$ is the Jordan product, then: (i) K_ω^J genuinely uses the Jordan structure (addressing F2 from the patch plan); (ii) the Fubini-Study distance $d_{FS}^J = \ell_0 \cdot \arccos(|K_\omega^J|)$ is still a metric (arccos on any inner-product space); (iii) the D=3 Gram-rank argument becomes a representation-theoretic question about Jordan algebra invariants.

Statement. Lemma (K_ω^J Jordan-built variant). If K_ω is redefined as $K_\omega^J(A,B) := \omega(A \circ B) / \sqrt{(\omega(A^\dagger A) \cdot \omega(B^\dagger B))}$ where $A \circ B = \frac{1}{2}(AB + BA)$ is the Jordan product, then: (i) K_ω^J genuinely uses the Jordan structure (addressing F2 from the patch plan); (ii) the Fubini-Study distance $d_{FS}^J = \ell_0 \cdot \arccos(|K_\omega^J|)$ is still a metric (arccos on any inner-product space); (iii) the D=3 Gram-rank argument becomes a representation-theoretic question about Jordan algebra invariants.

Proof sketch. (i) $K_\omega^J(A,B) = \omega(\frac{1}{2}(AB+BA))/\sqrt{(\dots)}$ uses the Jordan product directly, unlike the current $K_\omega = |\omega(A^\dagger B)|^2/\dots$ which is pure sesquilinear. (ii) K_ω^J is symmetric ($A \circ B = B \circ A$) and positive semi-definite (by the GNS construction), so $|K_\omega^J| \in [0,1]$ and $\arccos(|K_\omega^J|)$ is a valid metric by L1. (iii) The Gram matrix $G_{ij} = K_\omega^J(A_i, A_j)$ has rank determined by the number of independent symmetric bilinear invariants of the Jordan algebra A_J . If A_J supports exactly 3 independent invariants (a representation-theoretic claim), $\text{rank}(G) \leq 3$, giving D=3.

References. Jordan-von Neumann-Wigner (1934) classification of Jordan algebras.

Dependencies. L1. The representation-theoretic argument (iii) is HARD and needs separate development.

Summary of the Deep-Dive

Seven lemmas (L1–L7) formalize the Fubini-Study resolution. L1–L2 establish the metric and automatic triangle inequality (standard). L3 shows Factorization + Dominance become unnecessary. L4 establishes the log-as-potential relationship (the metric tensor is the Hessian of the log potential — this is the resolution of the user's -log vs arccos question). L5 reinterprets the cubic kernel as the Riemannian volume element. L6 shows κ_B and BH entropy survive the swap (ℓ_0 is algebraic). L7 addresses the K_ω^J Jordan-built variant. The chain $L1 \rightarrow L2 \rightarrow L3$ is standard and ready for proof execution. L4 is the key conceptual move. L6 is the ripple-effect guarantee. L7 is the hardest (representation-theoretic) and may require separate development.